

A Reproducibility

A.1 Experimental Settings

We evaluate each GLUE and SuperGLUE dataset using the metric specified in the benchmark. We reuse the evaluation code from the publicly available T5 open-source release to compute metrics.¹² For the SQuAD and MRQA datasets, we evaluate using F1, one of the metrics used by the SQuAD benchmark, where partial answer spans are considered. Again, we use the T5 open-source release for metric calculation.¹³ All of our models use T5 1.1 as the base frozen model, additional details and pre-trained checkpoints can be found on GitHub.^{14,15}

All prompts for T5 Small and Base models were trained on 4 TPU v2 chips, while prompts for larger models were trained on 16 TPU v3 chips.

Parameter counts for each prompt can be found in Table 4. Average runtimes until convergence can be found in Table 5.

A.2 Hyperparameter Search

This work used 77 hyperparameter search trials (40 for prompt tuning and 37 for single-task model tuning), and 3 training runs (with validation evaluation) for each baseline configuration and ablation setting, for a total of 195 runs for our main result and ablations. There were an additional 18 runs for the domain shift experiments and 24 extra runs to create the ensemble. Hyperparameter bounds can be found in Table 6. Hyperparameter tuning was done via manual tuning and settings were selected based on the SuperGLUE score. All experiments in this work, outside of the hyperparameter being ablated, use our [default configuration](#) of 100K steps of LM Adaption, a prompt length of 100, and “class-label” initialization.

All graphs of our experimental results plot the mean and standard deviation over 3 runs as computed by Seaborn (Waskom, 2021). Some settings have such low variance that the standard deviation

is hidden behind the line itself, such as “Model Tuning (Multi-task)” in Figure 1 and the Base, Large, and XL prompts trained on the “Span Corruption” pretraining objective in Figure 3(b). Figure 4 also shows mean and standard deviation for the number of parameters each method uses as the prompt length varies from 1–100. The “Prefix Tuning (Train)” curves appears to have no standard deviation because the parameter count is so strongly dominated by the cost of the reparameterization parameters that the standard deviation bands are occluded. For our experiments on domain transfer, we report mean and standard deviation over 3 runs.

A.3 Datasets

All datasets used are in English. For the GLUE^{16,17} and SuperGLUE¹⁸ datasets, we used the training, validation, and test splits that ship with TensorFlow Datasets. We used version 1.0.0 for GLUE and 1.0.2 for SuperGLUE datasets. For SQuAD¹⁹ we used v1.1:3.0.0 from Tensorflow Datasets and follow the provided training, validation, and test splits. For the out-of-domain datasets we used the development splits distributed as part of the MRQA shared task.²⁰ Dataset sizes can be found in Table 7. The label distributions for each dataset can be found in Table 8 (BoolQ), Table 9 (CB), Table 10 (COPA), Table 11 (MultiRC), Table 14 (RTE), Table 12 (WiC), Table 13 (WSC), Table 15 (MRPC) and Table 16 (QQP).

The question answering datasets are extractive datasets with a variety of answers, so there isn’t a label distribution to report. Similarly, the ReCoRD dataset is a multiple choice dataset where the model must predict the masked out entity from a list of possible entities. Due to this formulation there isn’t a meaningful label distribution.

We followed the open-source T5 preprocessing procedure²¹ for each dataset, except that we omit the dataset prefix denoting which SuperGLUE dataset an example belongs to. For the SQuAD and

¹²<https://github.com/google-research/text-to-text-transfer-transformer/blob/master/t5/evaluation/metrics.py>

¹³<https://github.com/google-research/text-to-text-transfer-transformer/blob/master/t5/evaluation/metrics.py#L151>

¹⁴https://github.com/google-research/text-to-text-transfer-transformer/blob/master/released_checkpoints.md#t511

¹⁵https://github.com/google-research/text-to-text-transfer-transformer/blob/main/released_checkpoints.md#lm-adapted-t511lm100k

¹⁶<https://www.tensorflow.org/datasets/catalog/glue#gluemrpc>

¹⁷<https://www.tensorflow.org/datasets/catalog/glue#glueqqp>

¹⁸https://www.tensorflow.org/datasets/catalog/super_glue

¹⁹https://www.tensorflow.org/datasets/catalog/squad#squadv11_default_config

²⁰<https://github.com/mrqa/MRQA-Shared-Task-2019#out-of-domain>

²¹<https://github.com/google-research/text-to-text-transfer-transformer/blob/master/t5/data/preprocessors.py>

T5 Size	Prompt Length	Trainable Parameters	Total Parameters	Percent Trainable
Small	1	512	76,961,664	0.00067%
	5	2,560	76,963,712	0.00333%
	20	10,420	76,971,572	0.01330%
	50	25,600	76,986,752	0.03325%
	100	51,200	77,012,352	0.06648%
	150	76,800	77,037,952	0.09969%
Base	1	768	247,578,624	0.00031%
	5	3,840	247,581,696	0.00155%
	20	15,360	247,593,216	0.00620%
	50	38,400	247,616,256	0.01551%
	100	76,800	247,654,656	0.03101%
	150	115,200	247,693,056	0.04651%
Large	1	1,024	783,151,104	0.00013%
	5	5,120	783,155,200	0.00065%
	20	20,480	783,170,560	0.00262%
	50	51,200	783,201,280	0.00654%
	100	102,400	783,252,480	0.01907%
	150	153,600	783,303,680	0.01961%
XL	1	2,048	2,849,759,232	0.00007%
	5	10,240	2,849,767,424	0.00036%
	20	40,960	2,849,798,144	0.00143%
	50	102,400	2,849,859,584	0.00359%
	100	204,800	2,849,961,984	0.00718%
	150	307,200	2,850,064,384	0.01078%
XXL	1	4,096	11,135,336,448	0.00004%
	5	20,480	11,135,352,832	0.00018%
	20	81,920	11,135,414,272	0.00074%
	50	204,800	11,137,380,352	0.00184%
	100	409,600	11,135,741,952	0.00368%
	150	614,400	11,135,946,752	0.00552%

Table 4: Number of parameters used for various prompt lengths and T5 model sizes. Trainable parameters is the number of parameters in the prompt itself, while total parameters includes the prompt plus the original T5 parameters. The T5 parameters are frozen and shared across all tasks, and include the SentencePiece lookup table parameters. The final column is the percentage of total parameters that are trainable.

MRQA datasets we used the T5 SQuAD preprocessing code²². By following the T5 preprocessing and text-to-text format, we recast the WSC dataset as a text generation task. Instead of predicting whether a supplied referent is correct for a highlighted span, our model predicts the correct referent directly. As such, we can only learn from training examples where the referent is correct, so WSC training data where the supplied referent is incorrect are omitted.

No new data was collected for this work.

Prompt Length	T5 Size	Time
1	Large	3:17 ±02:10
	XL	3:37 ±02:11
	XXL	21:23 ±01:54
20	XL	49:08 ±18:53
	XXL	53:03 ±16:25
50	Small	09:05 ±05:07
	Base	55:01 ±27:48
	Large	1:14:16 ±13:12
	XL	2:30:10 ±25:40
100	XXL	3:13:13 ±23:08
	Small	16:25 ±01:15
	Base	29:57 ±00:18
	Large	1:23:36 ±10:21
	XL	3:35:00 ±54:42
	XXL	3:51:15 ±45:53

Table 5: Mean and standard deviation of the runtime until convergence for the BoolQ dataset and various prompt lengths and model sizes. Convergence is defined as reaching a performance within 1% of the mean value for that model configuration. A few configurations have been omitted because their runtimes were artificially extended due to preemption.

²²<https://github.com/google-research/text-to-text-transfer-transformer/blob/master/t5/data/preprocessors.py#L264>

Hyperparameter	Search Space
Learning Rate	0.001–0.5
Parameter Scaling	{True, False}
Batch Size	{32, 64, 126, 256, 512}
Number of Steps	{10,000, 20,000, 30,000}
Warmup Steps	{off, 2,000, 3,000}
Decay Factor	{off, 0.1, 0.5}
Steps per Decay	{off, 4,000, 6,000, 8,000}

Table 6: Search space for each hyperparameter considered. Parameter Scaling refers to the Adafactor setting where an update is scaled by the norm of the parameter it will be applied to. Warmup Steps is the number of steps before a linearly increasing learning rate reaches the Learning Rate value, starting from zero. Decay Factor is the reduction in Learning Rate size that occurs every “Steps per Decay” steps.

Dataset	Training	Validation	Testing
BoolQ	9,427	3,270	3,245
CB	250	56	250
COPA	400	100	500
MultiRC	27,243	4,848	9,693
ReCoRD	100,730	10,000	10,000
RTE	2,490	277	3,000
WiC	5,428	638	1,400
WSC	259*	104	146
MRPC	3,668	408	1,725
QQP	363,849	40,430	390,965
SQuAD	87,599	10,570	-
TextbookQA	-	1,504	-
RACE	-	1,503	-
BioASQ	-	1,501	-
RE	-	674	-
DuoRC	-	2,948	-
DROP	-	1,503	-

Table 7: Sizes for training, validation, and testing splits of each dataset used. *Following T5, our casting of WSC as a text generation problems means we can only train on examples where the supplied referent is correct. This means our training dataset is smaller than the normal WSC training dataset, which has 554 examples.

Split	False	True
Training	37.7	62.3
Validation	37.8	62.2

Table 8: Label distribution for the BoolQ dataset.

Split	contradiction	entailment	neutral
Training	47.6	46.0	6.4
Validation	50.0	41.1	8.9

Table 9: Label distribution for the CB dataset.

Split	choice1	choice2
Training	48.8	51.2
Validation	55.0	45.0

Table 10: Label distribution for the COPA dataset.

Split	False	True
Training	55.9	44.1
Validation	57.2	42.8

Table 11: Label distribution for the MultiRC dataset.

Split	False	True
Training	50.0	50.0
Validation	50.0	50.0

Table 12: Label distribution for the WiC dataset.

Split	False	True
Training	0.0	100.0
Validation	63.5	36.5

Table 13: Label distribution for the WSC dataset. Following T5, we cast the WSC dataset to a free-form text generation task where the model generates the referent to the highlighted span instead predicting if the supplied entity is the correct referent of the highlighted span. Thus, we only use training data where the supplied referent is correct making our training label distribution focused entirely on True.

Split	entailment	not_entailment
Training	51.2	49.8
Validation	52.7	47.3

Table 14: Label distribution for the RTE dataset.

Split	equivalent	not_equivalent
Training	67.4	32.6
Validation	68.4	31.6

Table 15: Label distribution for the MRPC dataset.

Split	duplicate	not_duplicate
Training	36.9	63.1
Validation	36.8	63.2

Table 16: Label distribution for the QQP dataset.